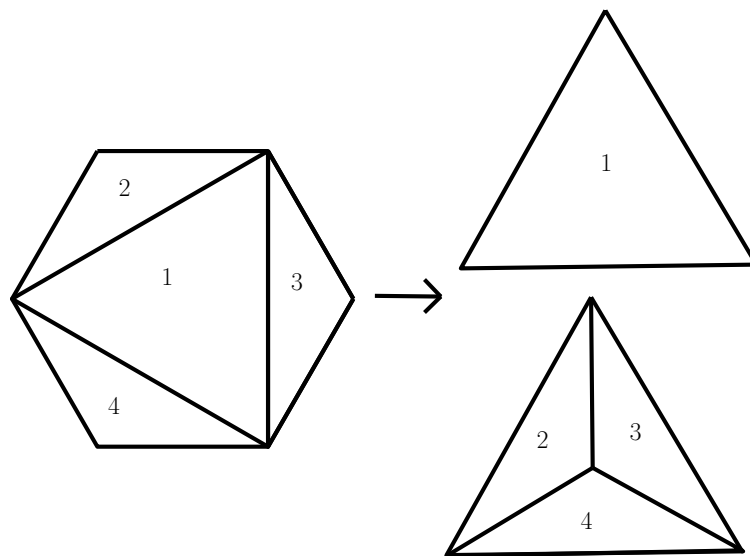


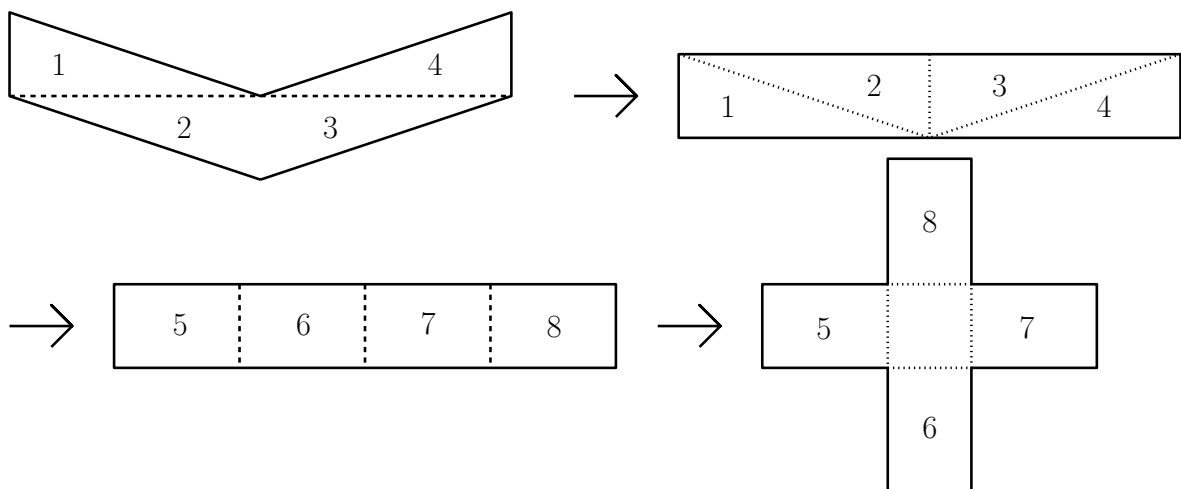
Dissection Puzzle Solutions

Ethan Sharma

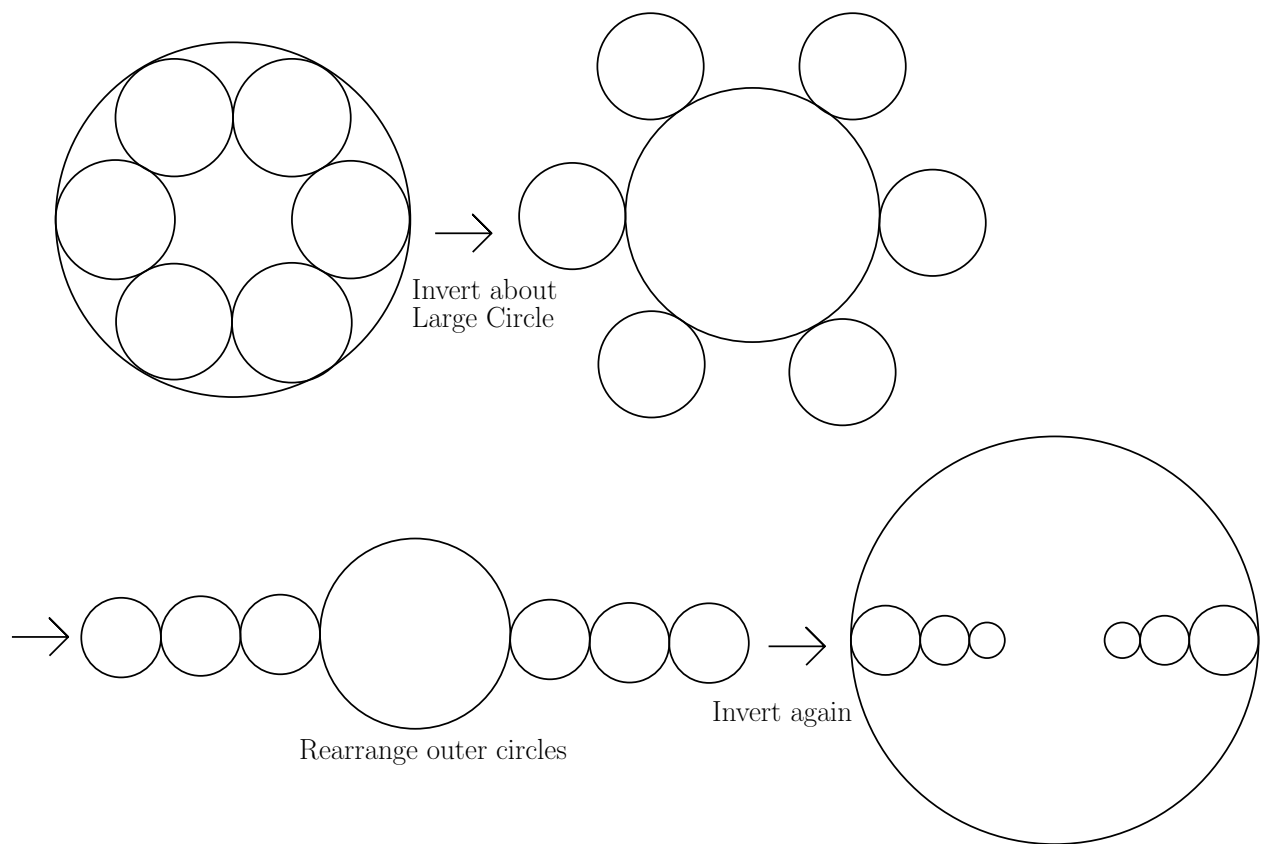
Problem 1



Problem 2

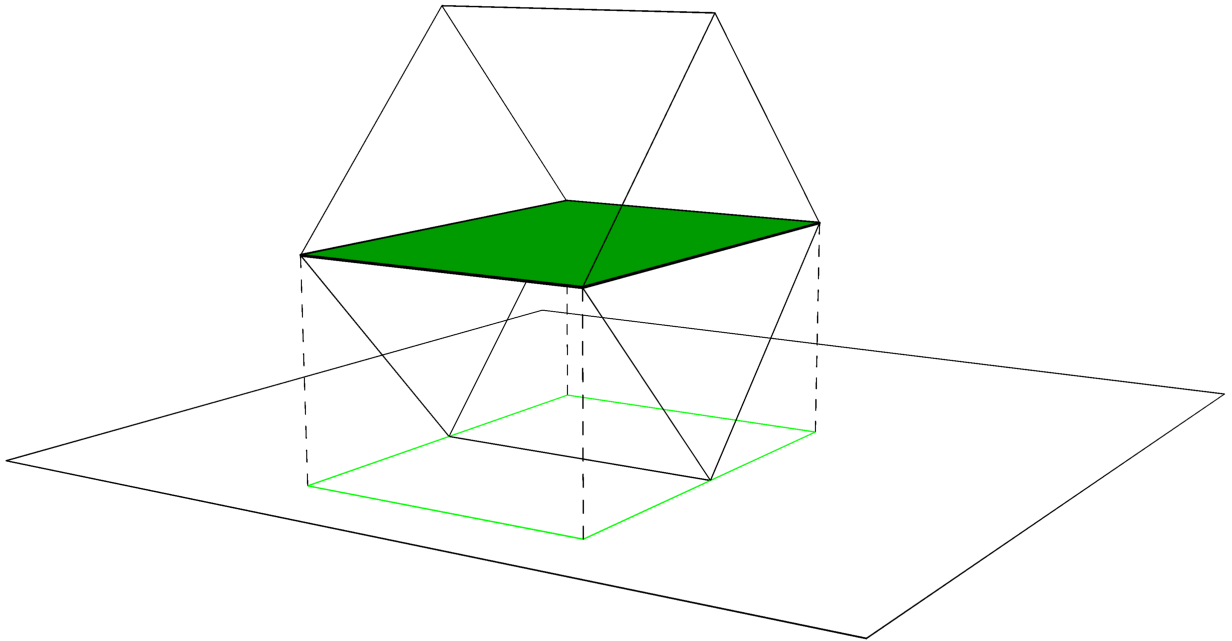


Problem 3



Problem 4

First, rotate the cube so that a rectangular cross section is perpendicular to the plane we plan on projecting onto. Alternatively, we can just choose the plane that'll give us a rectangular projection.



After projection, we are left with a rectangle with side lengths 1 and $\sqrt{2}$. We then cut along the main diagonal and rotate one piece, as shown, and we are left with a non-square kite.

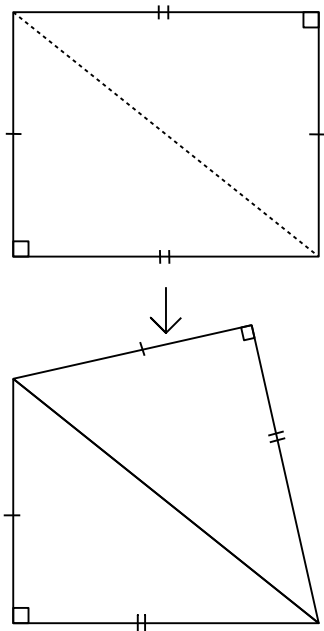


Figure 1: Rectangle to Non-Square Kite

Problem 5

We can perform cuts with two dimensional planes parallel to a face of the cube. This is a face view perpendicular to those planes to see the cuts. We can slightly bypass the requirement of cuts being full planes by just putting back together the dotted/dashed borders to reform the full cube.

Assume the initial cube is $5 \times 5 \times 5$ (for instructive purposes, it could be anything). The first four cuts is along the larger dashed lines, then we take out the $3 \times 3 \times 5$ rectangular prism out the middle and perform the four cuts along the dotted lines. Again, we put the edges back together and take out a $1 \times 1 \times 5$ rectangular prism. Then, we only show part of each prism to give an illusion of three cubes stacked on top of each other.

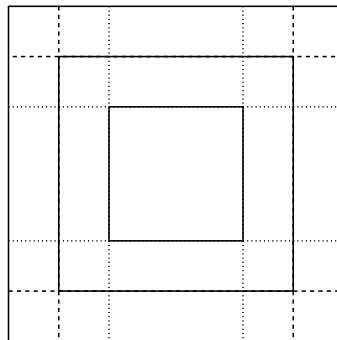


Figure 2: Face view. Dotted/Dashed lines are cut lines

Here is what the figure looks like with the insides shown:

